BREWERY OPERATIONS · LOGISTICS DIGITIZATION

TruckFlow

GATE MANAGEMENT PLATFORM

A mobile-first web application to digitize the full truck turnaround lifecycle — from gate entry to gate exit — replacing manual clipboard logging with real-time, QR-triggered stage tracking.

DOCUMENT TYPE	PHASE	STATUS
Product Requirements (PRD)	V1 — Web Demo (Approval)	Draft — For Review

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01 Problem Statement

CONTEXT

> Truck turnaround times at the brewery gate are currently tracked with pen and paper. There is no real-time visibility, no historical analytics, and no accountability when delays occur.

Every truck that enters the brewery passes through eight discrete operational stages, each with a measurable duration. Today, gate officers and yard staff record these transitions manually — filling out physical log sheets, sometimes missing timestamps entirely, and handing off paper forms between shifts. This creates several compounding problems:

PAIN POINT	CURRENT IMPACT
No real-time visibility	Supervisors cannot see where a truck is in the process without physically walking the yard
Manual time errors	Timestamps are written retrospectively, introducing inaccuracies of 5–30 minutes
No delay attribution	When a truck misses its slot, there is no data to identify which stage caused the bottleneck

Compliance gaps	Fueling and loading records cannot be audited reliably; handwriting is often illegible
No driver accountability	Driver arrival codes are verbal; disputes over arrival time cannot be resolved

02 Product Vision

DIRECTION

> TruckFlow replaces the clipboard with a smartphone. Every stage transition is captured digitally, in real time, with timestamps tied to the actor who triggered them.

8	~0s	100%
Tracked stages per truck visit	Target logging overhead per transition	Audit trail coverage (vs ~60% today)

SCOPE NOTE

This document covers **Version 1: a mobile-focused web application** built for demo and internal approval. The production-grade native mobile app (iOS/Android) is scoped to Version 2. All V1 design decisions must remain compatible with a native migration path.

03 Users & Roles

ACTORS

🚚 **Driver**	🛂 **Gate Officer**	⚙️ **Yard Operator**	⛽ **Fueling Staff**
📊 **Supervisor / Manager**	🔧 **System Admin**		

ROLE	PRIMARY INTERFACE	KEY PERMISSIONS
Driver	QR scan or code exchange at gate; receives status updates	Initiate Gate In only; view own truck status
Gate Officer	Mobile web app (phone or tablet at gatehouse)	Confirm Gate In, trigger Gate Out, view live queue
Yard Operator	Mobile web app (handheld in yard)	Log Unlatching, Offloading, Loading, Latching stage events
Fueling Staff	Mobile web app	Log Fueling start and end
Supervisor	Dashboard view (mobile or desktop)	View all active trucks, full history, analytics; no stage editing
Admin	Settings panel	Manage users, QR codes, truck registry, override records

04 The 8-Stage Gate Process

CORE MODEL

Each truck visit progresses through exactly eight sequential stages. The stages differ in how time is captured — some record only a single timestamp, one is derived automatically, and five capture an explicit start and end.

01 GATE IN → 02 QUEUING → 03 UNLATCH → 04 OFFLOAD → 05 LOADING → 06 LATCH

1 Gate In

Truck arrives at the brewery entrance. The gate officer or driver initiates entry via QR scan or manual code exchange. The exact arrival timestamp is recorded automatically at the

TIMESTAMP ONLY

QR OR CODE

moment of action.

2	**Queuing** The period between gate entry and when the unlatching bay becomes available. This duration is *automatically derived* — calculated as the difference between Gate In time and Unlatching Start time. No manual input is required.	AUTO-DERIVED
3	**Unlatching** The truck is brought to the bay and the trailer seal or latch is opened. Operator records the start and end of this operation.	START + END
4	**Offloading** Contents of the truck are offloaded (e.g. raw materials, returns, empties). Operator marks start when offloading begins and end when the truck is emptied and cleared.	START + END
5	**Loading** Outgoing product (finished goods, kegs, crates) is loaded onto the truck. Start is marked when loading commences; end is marked when the truck is fully loaded and the bay team signs off.	START + END
6	**Latching** The trailer or container is sealed, locked, and secured after loading. Operator records start when the latch process begins and end when the seal is confirmed closed.	START + END
7	**Fueling** The truck is fueled on-site. Fueling staff marks start when the nozzle connects and end when fueling is complete and the cap is sealed. Fuel volume may be optionally logged.	START + END

8	**Gate Out** The truck clears the exit gate. The gate officer confirms departure and the exact exit timestamp is captured. Total turnaround time is calculated as Gate Out minus Gate In.	TIMESTAMP ONLY

TIME CAPTURE SUMMARY

Exact timestamp: Stages 1 (Gate In) and 8 (Gate Out) — single tap captures the moment. · **Start + End duration:** Stages 3–7 — operator taps "Start" then "End" for each. · **Auto-derived:** Stage 2 (Queuing) — system calculates Gate In → Unlatch Start automatically.

05 Screen Specifications

UX

All screens are designed mobile-first (375px base) with a focus on single-handed operation and outdoor readability (high contrast, large tap targets ≥ 48px).

SCR-01 — Gate In / Truck Registration

ROLE: DRIVER · GATE OFFICER

- → Two entry modes selectable at top: **[Scan QR Code]** or **[Enter Code Manually]**
- → QR mode opens device camera; scanning a valid truck QR code auto-populates truck ID and driver name
- → Manual mode shows a large numeric keypad / text field for code entry; gate officer can confirm on behalf of driver
- → On confirmation, timestamp is recorded server-side instantly — not when the form was opened
- → Success screen shows: Truck ID, Driver Name, Gate In Time, and assigned Queue number
- → System notifies yard operations panel of new truck in queue

SCR-02 — Live Queue Board ROLE: GATE OFFICER · YARD OPERATOR · SUPERVISOR

- → Shows all trucks currently on-site, sorted by Gate In time (oldest first)
- → Each truck card displays: Truck ID, current stage (highlighted), time in current stage, total time on-site
- → Trucks awaiting bay are marked as QUEUING with a live elapsed timer
- → Color coding: green = on track, amber = approaching threshold, red = exceeded expected duration
- → Tapping a truck card opens the Stage Action view for that truck
- → Supervisor sees a read-only version with no action buttons

SCR-03 — Stage Action View (Stages 3–7) ROLE: YARD OPERATOR · FUELING STAFF

- → Large, full-screen stage card showing: Truck ID, current stage name, stage number (e.g. "3 of 8")
- → Two large action buttons: **[START]** (green) and **[END]** (red) — END is disabled until START is tapped
- → Once START is tapped, a live stopwatch runs on-screen; server records start timestamp
- → Tapping END records end timestamp and automatically advances the truck to the next stage
- → Optional notes field per stage (e.g. "Loading halted — forklift issue") for exception logging
- → Confirmation modal before END to prevent accidental taps

SCR-04 — Gate Out ROLE: GATE OFFICER

- → Triggered when truck reaches Stage 8; gate officer taps truck in queue or scans QR at exit

- → Single large **[CONFIRM GATE OUT]** button with truck details displayed
- → On confirmation: records exact departure timestamp, calculates total turnaround time
- → Summary card shown: all 8 stage durations, total time, any flagged overruns
- → Option to share/export the trip summary as PDF or forward to driver WhatsApp (V2)

SCR-05 — Supervisor Dashboard ROLE: SUPERVISOR · MANAGER

- → Summary metrics row: trucks on-site today, completed today, average turnaround time, longest active truck
- → Live truck table (auto-refreshing) with all active visits
- → Historical log: searchable/filterable by date, driver, truck ID, stage
- → Stage duration analytics: bar chart of average time per stage over selected period
- → Bottleneck flag: highlights which stage accounts for the most delay across all visits

06 Functional Requirements

MUST / SHOULD / COULD

ID	REQUIREMENT	PRIORITY	NOTES
FR-01	Gate In via QR code scan using device camera	MUST	Camera access via browser (getUserMedia)
FR-02	Gate In via manual alphanumeric code entry as fallback	MUST	For when driver has no QR, or camera unavailable
FR-03	Server-side timestamping for all stage events (not client clock)	MUST	Prevents manipulation; single

			source of truth
FR-04	Stage 2 (Queuing) duration auto-calculated	MUST	= Unlatch Start – Gate In
FR-05	Start/End capture for Stages 3–7 with live stopwatch	MUST	START disables until tapped; END disabled until START
FR-06	Role-based access: gate officer, yard operator, fueling, supervisor, admin	MUST	Login with phone number + PIN (V1 simplicity)
FR-07	Live queue board visible to all staff roles	MUST	Auto-refresh every 10–15s or via WebSocket
FR-08	Full visit history per truck: all 8 stages with timestamps	MUST	Core audit trail requirement
FR-09	Optional notes field per stage for exception logging	SHOULD	Delays, incidents, equipment issues
FR-10	Configurable duration thresholds per stage (alert when exceeded)	SHOULD	e.g. Fueling > 30 min triggers amber flag
FR-11	Supervisor analytics dashboard with stage breakdown charts	SHOULD	Average durations per stage over time
FR-12	CSV / PDF export of visit records	SHOULD	For compliance and operations reporting
FR-13	Push notifications to relevant role when a truck enters their stage	COULD	Web push notifications (V1 nice-to-have)
FR-14	WhatsApp integration for driver visit summary on Gate Out	COULD	Deferred to V2
FR-15	Offline mode with local queue and sync on reconnect	COULD	Important for spotty yard connectivity —

			V2

07 Non-Functional Requirements

QUALITY

CATEGORY	REQUIREMENT
Performance	Stage action buttons must respond in under 300ms. Queue board must load in under 2s on 3G.
Usability	All primary actions must be reachable with one thumb. No critical action should require more than 2 taps from the queue board. Tap targets minimum 48×48px.
Accessibility	Outdoor readability: minimum 4.5:1 contrast ratio on all interactive elements. Font sizes ≥ 16px for body text.
Security	All timestamps stored server-side; client cannot submit an arbitrary time. Role permissions enforced server-side, not client-side.
Reliability	System must not lose a stage timestamp due to network timeout — queue locally and retry. Target 99.5% uptime.
Auditability	Every record must log which user account triggered it, the device, and the server-side timestamp. Records are immutable; corrections create new entries with a linked override flag.
Compatibility	Must function on Android Chrome (primary) and Safari iOS. No app install required for V1.

08 Data Model (Overview)

BACKEND

The following tables represent the core data entities required. Schema design to

be finalized in the technical spec.

ENTITY	KEY FIELDS
Truck	`truck_id, plate_number, driver_name, qr_code_ref, status`
Visit	`visit_id, truck_id, gate_in_at, gate_out_at, total_duration_mins, current_stage`
StageEvent	`event_id, visit_id, stage_num (1–8), start_at, end_at, duration_mins, triggered_by_user_id, notes, is_override`
User	`user_id, phone, pin_hash, role, full_name, is_active`
QRCode	`qr_id, truck_id, code_string, issued_at, expires_at, is_single_use`

DERIVED FIELD

Queuing duration is never stored as a raw input. It is computed on read as: `stage_3.start_at − visit.gate_in_at`. This ensures it stays accurate even if Gate In is corrected via an admin override.

09 Roadmap & Phasing

PLANNING

PHASE	DELIVERABLE	GOAL
V1 — Demo Current scope	Mobile-focused web app covering all 8 stages, QR/code Gate In, live queue, supervisor view	Prove the concept to stakeholders; get approval for full build
V2 — Production Web	Full role system, analytics dashboard, CSV export, admin controls, offline resilience, push notifications	Replace paper entirely; full operations deployment
V3 — Native	iOS + Android native apps, WhatsApp integration, NFC tag support at bays,	Optimal UX for yard staff; deeper

App	driver self-service portal	integrations with fleet systems

10 Open Questions

DECISIONS NEEDED

#	QUESTION	OWNER
OQ-01	Can stages 3–7 ever run in parallel, or is the sequence always strictly linear? (e.g. can fueling start while latching is in progress?)	Operations Lead
OQ-02	Are QR codes per-truck (permanent) or per-visit (single-use, generated per trip)? Single-use is more secure but requires a printing or sharing mechanism.	Gate Operations
OQ-03	What is the expected peak number of trucks on-site simultaneously? This affects real-time sync architecture choices.	Ops Manager
OQ-04	Who handles admin override of an incorrect timestamp? Is this gate officer level or supervisor only?	Compliance
OQ-05	Should the system send any external notifications (SMS, email) to the driver or dispatch office on Gate Out?	Logistics Team
OQ-06	Is fueling always done on-site? If some trucks fuel off-site, should Stage 7 be skippable?	Operations Lead